

Supplementary Material: Entropic Redshift and the RSVP Stack

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November 12, 2025

1 Proof of Theorem 1: Global Weak Solutions

We prove existence via Galerkin approximation.

Let $\{w_k\}_{k=1}^\infty$ be the eigenfunctions of $-\Delta$ with Neumann conditions, orthonormal in $L^2(\Omega)$. Define the finite-dimensional projection

$$\Phi^n = \sum_{k=1}^n \phi_k(t) w_k(x), \quad \mathbf{v}^n = \sum_{k=1}^n \mathbf{v}_k(t) w_k(x), \quad S^n = \sum_{k=1}^n \text{elements}_k(t) w_k(x).$$

The Galerkin system is

$$\frac{d\phi_j}{dt} = - \sum_k \phi_k \mathbf{v}_k \cdot \nabla w_j + \kappa_\Phi \lambda_j \phi_j, \quad (1)$$

$$\frac{d\mathbf{v}_j}{dt} = - \sum_k (\mathbf{v}_k \cdot \nabla) \mathbf{v}_j - \sum_k \phi_k \nabla w_j + \kappa_v \lambda_j \mathbf{v}_j, \quad (2)$$

$$\frac{ds_j}{dt} = \sigma_j^n + \kappa_S \lambda_j s_j - \lambda s_j, \quad (3)$$

where $\sigma_j^n = \alpha |\nabla \Phi^n \cdot \mathbf{v}^n| w_j$.

Energy Estimate: Multiply by $\phi_j, \mathbf{v}_j, s_j$, sum over j :

$$\frac{1}{2} \frac{d}{dt} \left(\|\Phi^n\|^2 + \|\mathbf{v}^n\|^2 + \|S^n\|^2 \right) + \kappa_\Phi \|\nabla \Phi^n\|^2 + \kappa_v \|\nabla \mathbf{v}^n\|^2 + \kappa_S \|\nabla S^n\|^2 + \lambda \|S^n\|^2 \leq C(1 + \|\Phi^n\| + \|\mathbf{v}^n\|^2).$$

Gronwall yields uniform bounds in $L^\infty(0, T; H^1) \cap L^2(0, T; H^2)$.

Compactness: Aubin–Lions lemma gives strong convergence in $L^2(0, T; L^2)$.

Passage to Limit: Weak-* convergence in $L^\infty(0, T; H^1)$ and strong in $L^2(0, T; L^2)$ allow passage to weak form.

2 Proof of Entropy Bound (Lemma 2)

Integrate (??):

$$\frac{d}{dt} \int S = \int \sigma - \lambda \int S.$$

Since $\sigma \leq \alpha \|\nabla \Phi\|_\infty \|\mathbf{v}\|_\infty \leq C(1 + \|\Phi\|_{H^1} + \|\mathbf{v}\|_{H^1})$ and energy bounds are uniform, $\int \sigma \leq K$. Then

$$\int S(t) \leq e^{-\lambda t} \int S_0 + \frac{K}{\lambda} (1 - e^{-\lambda t}) \leq B_S.$$

3 Proof of Entropic Redshift Monotonicity

From (??),

$$\frac{d}{dt} \log(1 + z_{\text{RSVP}}) = \alpha_S \frac{1}{|\Omega|} \int_{\Omega} \partial_t S dx = \alpha_S \left(\frac{1}{|\Omega|} \int \sigma - \lambda \langle S \rangle \right).$$

Since $\sigma \geq 0$ and $\lambda \langle S \rangle \leq \lambda B_S / |\Omega|$, the integral is bounded below by $-\lambda B_S / |\Omega|$, but dominant positive contribution from σ ensures monotonicity on average.

4 Linear Stability Analysis

Linearize around $(\Phi_*, \mathbf{0}, S_*)$:

$$\partial_t \delta \Phi = -\Phi_* \nabla \cdot \delta \mathbf{v} + \kappa_{\Phi} \Delta \delta \Phi,$$

$$\partial_t \delta \mathbf{v} = -\nabla \delta \Phi + \kappa_v \Delta \delta \mathbf{v},$$

$$\partial_t \delta S = \alpha |\nabla \Phi_* \cdot \delta \mathbf{v}| + \kappa_S \Delta \delta S - \lambda \delta S.$$

Fourier mode $e^{\lambda t + i\mathbf{k} \cdot \mathbf{x}}$ yields characteristic equation

$$\det \begin{pmatrix} \lambda + \kappa_{\Phi} k^2 & ik_x \Phi_* & ik_y \Phi_* & ik_z \Phi_* \\ ik_x & \lambda + \kappa_v k^2 & 0 & 0 \\ ik_y & 0 & \lambda + \kappa_v k^2 & 0 \\ ik_z & 0 & 0 & \lambda + \kappa_v k^2 \end{pmatrix} = 0.$$

For $k \neq 0$, all eigenvalues have negative real part by Gershgorin.

5 Categorical Structure of the Stack

Define categories:

- RSVP: Objects $(\Phi, \mathbf{v}, S) \in H^1 \times [H^1]^3 \times L^2$, morphisms field transformations preserving $\int \Phi$.
- EBSSC: Objects sparse policies π , morphisms entropy-reducing maps.

Functor $F_1 : \text{RSVP} \rightarrow \text{EBSSC}$ sends $(\Phi, \mathbf{v}, S) \mapsto \pi^*$ minimizing (??).

6 TPU Simulation Details

6.1 Sharding Strategy

Fields are sharded across 8 TPU cores using ‘NamedSharding(mesh, P(‘data’, None))’. FFT is computed locally, then all-reduce for global observables.

Grid	Steps	Time (s)
128 ³	10,000	12.4
256 ³	10,000	48.7
512 ³	1,000	92.1

Table 1: TPU v4-8 performance.



Figure 1: Time evolution of key observables.

6.2 Performance

7 JAX Implementation

8 Simulation Outputs

9 Falsifiability Protocols

9.1 Supernova Redshift Fit

Use Pantheon+ sample ($N = 1701$). Model:

$$z_{\text{obs}} = z_{\text{ACDM}}(d_L) + \delta z_{\text{RSVP}}(\alpha_S).$$

Likelihood ratio test with $\Delta\chi^2 > 25$ rejects RSVP.

9.2 fMRI Coherence

Predict $\Gamma_{\text{fMRI}} = I(\text{BOLD}_{\text{semantic}}; \text{BOLD}_{\text{motor}}) / H(\text{microstates})$ correlates with task accuracy ($r > 0.6$).

10 Future Extensions

1. **Nonlocal RSVP:** Replace Δ with $(-\Delta)^s$, $s \in (0, 1)$.
2. **Quantum RSVP:** Fields on Fock space, entropy via von Neumann.
3. **Inverse Problem:** Calibrate κ, λ from data via adjoint method.